

Plan 08 v0: Learned Memory Wrapper for Long-Context RCA

Progress / Fact Record

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RCA Foundation Model / Long-Context Memory

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Combined progress / fact record
Weekly inputs are appended below in chronological order.

Plan 08 / Paper B — June Week 5

Mon 2026-06-29 to Sun 2026-07-05

Leader update: research outcome, long-context facts, v2.0.0 options, and resource ask.

Scientific Question

Main question

When a model faces long evidence, what should it keep, what should it ignore, and how can it know whether its reduced context is still reliable?

Why this is scientific, not only engineering

“Long context” is not one capability. It decomposes into retrieval, local extraction, distributed semantic reasoning, distractor robustness, and architecture-dependent memory access.

- The core object of study is the **interaction between task type, context length, and reduction method**.
- The scientific goal is to map this interaction, then design a method that adapts instead of assuming one fixed rule.

Background: What Did We Test?

Benchmarks = task regimes

- **RULER / NIAH**: synthetic needle retrieval; controls length.
- **numerical / categorical / coding NIAH**: exact value, label, or code-style retrieval.
- **SQuAD v2**: local extractive QA; tests whether a recent window is enough.
- **HotpotQA**: multi-hop QA; evidence is distributed.
- **NarrativeQA / QuALITY**: long, abstractive/global reading; tests focus under distractors.
- **TriviaQA / LoCoMo / MuSR**: noisy evidence, dialogue memory, and reasoning variants.

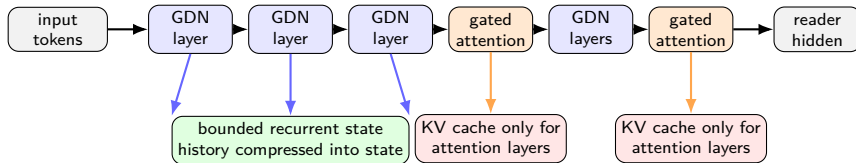
Baselines = context-reduction strategies

- **full / no-context**: ceiling and floor.
- **window / truncation**: assumes answer is local or recent.
- **RAG / BM25**: assumes query and evidence share lexical overlap.
- **Prompt compression (LLMLingua)**: assumes token salience predicts usefulness.
- **KV eviction**: SnapKV/H2O/Expected/TOVA, Streaming, Knorm, KVzip/FastKVzip.
- **Token merging**: tests whether similar tokens/spans can be merged without losing the answer.

Readout

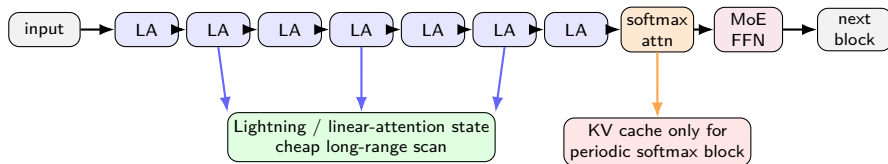
The experiment matrix asks which method fails in which task regime, not only which method has the best average score.

Background: Qwen3.5 / 3.6 Hybrid GDN Stack



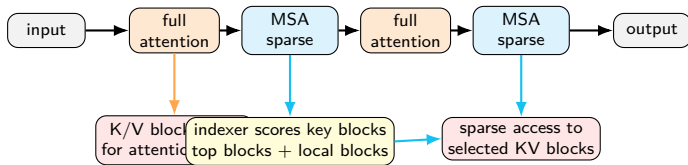
Prefill dependency: most layers scan tokens into a GDN recurrent state; periodic attention layers create ordinary K/V for exact lookup. **KV-cache methods apply only to the attention subset, not to GDN layers.** Qwen3.6 is reported as KV on 16/64 layers, so the KV lever is partial.

Background: MiniMax-Text-01 / M1 Hybrid Lightning Stack



Prefill dependency: the published MiniMax-Text-01/M1 pattern is roughly **7 Lightning/linear-attention blocks + 1 softmax-attention block**, with MoE capacity. Lightning blocks carry cheap long-range state; softmax blocks periodically restore exact token interactions. **KV eviction applies only to the softmax blocks.**

Background: MiniMax-M3 Sparse-Attention Stack



Prefill dependency: M3 is not recurrent/linear. It is a sparse-attention Transformer: sparse layers use an indexer to choose key blocks, then exact attention runs on those selected blocks. **KV exists, but access is already learned and sparse;** KV eviction methods are possible only as a comparison to MSA, not as the native mechanism.

Experimental Facts We Established

- **Fact 1: fixed baselines are regime-specific.** Window, RAG, prompt compression, and KV eviction each work in some regimes and fail in others.
- **Fact 2: length changes method ranking.** Attention-based KV eviction works around shorter contexts but can collapse at 16k+.
- **Fact 3: task type changes method ranking.** Retrieval, extractive QA, global reasoning, and literary MC prefer different context reduction behavior.
- **Fact 4: more context is not always better.** Full context can hurt when the context introduces distractors or exceeds the model's usable reasoning ability.
- **Fact 5: model architecture matters.** Linear-attention / GDN-style bases do not expose the same KV-cache lever as dense-attention models.

Current Paper Intro: Story We Can Tell

Intro logic

Modern LLMs can accept longer contexts, but long-context use is not solved: full context is costly and sometimes harmful; simple reduction methods are brittle; and different tasks require different evidence-preservation strategies.

- Start with the field assumption: longer windows and context compression should help.
- Show the contradiction: the same method can win on retrieval and lose on semantic QA; full context can even hurt.
- Therefore the missing piece is not “one better compressor” but a mechanism that can **adapt to the evidence regime**.
- This motivates learned memory plus a reliability decision.

What v1.8.0 Already Gives Us

- A working learned-memory path on high-headroom tasks after fixing termination supervision and joint answer signal.
- A gate/fallback formulation that is useful conceptually: use memory when reliable, fall back when unsafe.
- Ablation evidence that the carrier is not fragile: many knobs move the compressed path by only a few points.
- A clearer negative result: the current gate story must become held-out or conformal, not in-sample threshold picking.

Takeaway

v1.8.0 is the prototype and evidence base. v2.0.0 should sharpen the claim into a cleaner reliability theory and a better necessity regime.

Next Paper Initialization Direction

Working title direction

Adaptive Evidence Memory for Long-Context Reasoning

- **Paper role of the fact-base:** establish that fixed black-box context reduction is not enough.
- **Paper role of the method:** introduce a memory that preserves evidence and a reliability signal that decides when to trust it.
- **Core novelty target:** self-verified memory + calibrated routing, not just another compressor.
- **Evaluation target:** public long-context regimes where window, RAG, prompt compression, and KV eviction fail for different reasons.

- ① **Self-verified evidence memory:** train the memory to preserve evidence; use that preservation/reconstruction signal as the reliability check.
- ② **Reliability-gated fallback:** use extracted knowledge when safe; otherwise fall back to raw context, retrieval, or a higher-cost path.
- ③ **Hybrid evidence memory:** combine a raw recent window with compressed global memory.
- ④ **Adaptive budget:** easy contexts get fewer memory tokens; dense or ambiguous contexts get more.

Next Experiments To Initialize

- **M1: Hybrid Evidence Memory** — reader sees raw recent window + compressed global memory.
- **M2: Conformal router** — held-out reliability calibration instead of in-sample threshold picking.
- **M3: Long-context necessity wave** — semantic-spread tasks plus retrieval honest-negative.
- **M4: Transfer matrix** — train on one domain, test on another; measure whether memory is domain-invariant.
- **M5: Figures** — turn the fact-base into length, task, and method-failure figures.

What Leaders Need to Decide

Option A: Fact-base paper Publish the long-context baseline map first; method comes later.

Option B: Method paper Build v2.0.0 now around self-verified evidence memory + conformal routing.

Option C: Two-stage plan Fact-base as Section 1 / motivation, then v2.0.0 as the method contribution.

Recommendation

Option C: use the fact-base to define the gap, then spend compute only on the minimum v2.0.0 experiments needed to show the gap can be closed.

Recommended Next Step

- Freeze the public long-context fact-base and produce the figures.
- Implement Hybrid Evidence Memory: raw window + compressed memory.
- Add conformal reliability routing and report risk-coverage / cost-coverage.
- Run the 5x5 transfer matrix to test cross-domain memory generalization.
- Decide whether v2.0.0 is a full method paper or a fact-base + position paper.

Ask

Approve GPU budget for the v2.0.0 public benchmark experiments, especially the hybrid-memory and transfer-matrix runs.

Plan 08 / Long-Context Compression — July Week 1

Mon 2026-07-06 to Sun 2026-07-12

Focus: **Paper B (IMP)**; recall of Paper A. High-level logic here; details linked.

Details hub: [OVERVIEW](#) · [fact-base](#) · [matrix-facts](#)

Recall — the problem and the two papers

Problem (why it matters)

LLMs accept long inputs, but long context is not solved: full context is costly and sometimes *hurts*; training-free reduction methods are brittle; different tasks need different evidence.

Paper A compression **robustness via per-input validity-detection** — trust a compressed memory only when a self-verification signal says it's safe, else fall back to full (a “do-no-harm gate”).

Paper B **observe long-context failures, then attach a lightweight importance-routing structure to a frozen base** (plug-and-play or lightly trained). *This week's focus.*

Full framing: [OVERVIEW §2](#).

Preliminaries — terms (plain language)

- **Long-context / needle:** long input; the “needle” is the small fact that answers the question.
- **Frozen base:** pretrained LLM, weights unchanged — we add things around it, never retrain.
- **KV cache / KV-eviction:** attention’s per-token memory / dropping cached tokens. **KV-free compression:** reduce the input itself (prune / merge / retrieve) — works on any architecture incl. cache-free linear models.
- **full / no_ctx:** references — whole input vs. no document (ceiling / floor).
- **Importance routing (IMP):** score token importance, keep the important, drop the rest, before the base reads.
- **query-relevance / surprisal:** how related to the question / how unexpected a token is.
- **span vs token:** keep whole contiguous spans vs. isolated tokens. **do-no-harm gate:** per-input trust-or-fallback.

Paper A — recall + a clean negative this week

Thesis

Robustness via a validity detector (gate): trust the cheap compressed memory only when safe; else fall back to full.

- **Reconstruction-as-gate rejected (honest):** a random-memory control shows the signal $\approx$ noise across a 15-cell ablation and a high-K bracket; the earlier “+6pt” was noise.
- **Root cause:** the learned soft-memory **can't compress extractive QA** — 4 recipes all give compress $\approx$ no_ctx (≤ 0.20 vs full 0.69). Nothing to gate on.
- **Consequence:** ship the *confidence* gate; reconstruction blocked on a better compressor. Effort tilts to Paper B.

Detail: [decisions D24–D26](#) · [fact-base §12.3](#).

Paper B — observation $\rightarrow$ method

Observations driving the design

(1) No universal baseline; (2) naive merging kills the needle (ToMe = 0 on retrieval); (3) a cheap $O(L)$ signal finds the needle (query-relevance / surprisal); (4) attention-free is length-robust.

Method (IMP-v2.1.0, span-level, Mode A)

All results = IMP-v2.1.0 (span-32, keep-0.5, signals query+surprisal); token-level v2.0 superseded.

Score each token by cheap signals $\Rightarrow$ **keep the top- p important spans verbatim**, drop the rest $\Rightarrow$ frozen base reads the short sequence. Modes: **plug-and-play** (no training) / **light-weight** (tiny distilled router + side-cache for linear models).

Full method + implementation: [imp-method-and-implementation](#) · spec: [PAPER-B](#).

Paper B — results (high level)

- **Retrieval = full** at every length incl. 16k (where ToMe = 0, SnapKV collapses).
- **Token-level shreds coherent QA; span-level rescues it** (squad 0.15→0.46, hotpot 0.21→0.42) while keeping retrieval — the key ablation.
- **Beats full by denoising** on trivia and categorical; keep-rate curve is monotone.
- **Real-length LongBench (32k, no truncation)**: on real long docs the base scores low and **compression often beats full** (necessity regime) — stronger than synthetic NIAH.

All numbers (main table, token-vs-span, keep curve, LongBench): [fact-base §12–§12.3](#) · figures + narrative: [OVERVIEW §3.1](#).

Analysis — who wins where, and IMP's honest standing

- **Retrieval:** kvzip / criticalkv / LLMLingua / RAG / IMP all $\approx$ full; ToMe/SnapKV/H2O fail.
- **Extractive/multi-hop QA:** **RAG is the strongest KV-free baseline**; window is a surprisingly strong trivial baseline when the answer is local.
- **IMP's honest position:** matches full on retrieval and recovers QA, but on *short-context* extractive QA it does **not yet beat RAG / LLMLingua**. Its distinctive value is **training-free** · **architecture-agnostic (incl. cache-free linear)** · **denoising** · **real-length wins** — not raw short-context accuracy.

Snapshot main table (all methods $\times$ benchmarks): [fact-base §12](#).

Limitations & critical review (honest)

- No short-context accuracy win over RAG/LLMLingua yet; the case rests on generality + efficiency + long-context, which we must *show*.
- Prefill-saving is conditional (surprisal needs a full forward on quadratic bases unless a small scorer is used).
- **Not yet run on a linear/GDN base** — the one setting KV methods can't touch and IMP is uniquely useful.
- Equal-weight signal is sub-optimal; learned router (Mode B) unbuilt. narrativeqa/musr near base ceiling. Paper A blocked on the compressor.

Critical question

“Is IMP just a worse RAG?” — only defensible if we show (a) it works on cache-free linear models, (b) it wins in the necessity regime (input overruns the window), (c) semantic importance helps where lexical BM25-RAG fails. **These are the make-or-break experiments.**

Status & Next

Running a 12h / 6-GPU grand grid: full method $\times$ benchmark main table + all ablations (details land in the fact-base tomorrow).

Next plan (make-or-break first)

- ① **Linear/GDN base** — IMP where KV methods don't exist (unique territory).
- ② **Necessity regime** — inputs overrunning the window (LongBench / ∞ Bench $> 100k$).
- ③ **Semantic $>$ lexical** — help on abstractive inputs where BM25-RAG fails.
- ④ Then: small draft-LM scorer (real prefill saving) $\cdot$ learned router (Mode B) $\cdot$ harvest grand grid $\rightarrow$ full table + figures.

Ask

Endorse the pivot to **Paper B (IMP)** as the primary line; keep Paper A's confidence-gate as secondary until a better compressor exists.

Artifacts & detailed data (all linked)

- [OVERVIEW-both-papers-and-facts.md](#) — narrative, figures, both papers, §3.1 new experiments.
- [baseline-factbase-v2.0.0.md](#) — every number: KV-free family (§10–11), IMP main table + ablations (§12), Paper-A compressor negative (§12.3).
- [matrix-facts.md](#) — one-line facts F1–F26 with evidence pointers.
- [imp-method-and-implementation.md](#) — IMP algorithm + exact code path.
- [PAPER-B-v2.1.0-complete.md](#) · [PAPER-A-v1.8.1-complete.md](#) — paper specs.
- [decisions-2026-06-24.md](#) — dated decision log (D1–D26).

Plan 08 / Long-Context Compression — July Week 2

Mon 2026-07-13 to Sun 2026-07-19

This week: **Paper B rigor pass** (bug fix, corrected results, Occam v3.0) · **Paper C** (linear attention) opened · research deck.

First: a **full-project recall** (whole cycle, not just last week).

Recall (1/3) — background: the long-context memory wall

Why the project exists

LLMs accept long inputs, but long context is *not* solved: softmax attention keeps a **growing KV cache** (memory $\propto$ length), full context is costly and sometimes *hurts* accuracy, and training-free reduction methods are brittle and task-dependent.

- Two escape routes the field takes: **shrink the cache** (KV-eviction / prompt-compression / retrieval) vs. **remove the cache** (**linear attention** / SSM: a fixed-size recurrent state, $O(L)$ time, $O(1)$ memory).
- Our bases span both: **Qwen3** (quadratic) and **Qwen3.5/3.6** (Gated-DeltaNet linear 3:1 hybrid).

Guiding question (whole project)

On a *frozen* base, how do we feed / hold the *right* context so long-context accuracy survives — across architectures?

Recall (2/3) — one thesis, three papers (the whole cycle)

Thesis

Long-context accuracy is a **keep-the-answer-span** problem. No fixed compressor wins everywhere; the right mechanism is **input-adaptive** and ideally **training-free & architecture-agnostic**.

Paper A Learned **soft-memory** compressor on a frozen base + a do-no-harm gate. $\Rightarrow$ **honest negative**: compressor is a commodity, gate a non-effect, can't compress extractive QA. *This motivated the pivot.*

Paper B Stop building a compressor; **diagnose** what works + attach a training-free **importance-routing prefilter (IMP)** to a frozen base. *Main result line.*

Paper C (v3.0) The diagnosis reproduces on **cache-free linear** models where KV-methods can't run $\Rightarrow$ new line on **linear-attention forgetting / memory-state** (Qwen3.5/3.6 GDN family).

Deck: [main-research](#) · framing: [OVERVIEW](#).

Recall (3/3) — what each paper established (to date)

Paper A Learned compressor $\approx$ no-context on extractive QA (4 recipes); reconstruction-gate $\approx$ chance; **training-free IMP beats the learned compressor**. $\Rightarrow$ blocked; pivot justified. [negatives](#)

Paper B — diagnosis Full-test map over **9 model families**: (i) no universal compressor; (ii) “more context can hurt” (literary-MC full < blind); (iii) retrieval $\gg$ full on ultra-long; (iv) **model-invariant**, incl. linear GDN. [facts F27/F31/F43](#)

Paper B — method **IMP** = training-free, input-side, arch-agnostic router; keeps top spans/chunks verbatim; routes granularity per input. [method](#)

Paper C Landscape mapped (delta-rule family: DeltaNet $\rightarrow$ GDN $\rightarrow$ KDA $\rightarrow$ GDN-2); the linear **recall/forgetting bottleneck** is the opening. [research line](#)

This week (1/3) — Paper B: a rigor pass

- **Found & fixed a bug (F44).** IMP was *truncating* context to 16k *before* selecting, while RAG saw the whole 131k-token doc — 87% of RAG's kept content lived beyond 16k. Fix (full-doc retrieval): ∞ Bench 51 $\rightarrow$ **76**, the -19 gap *reversed* to a win. [audit](#)
- **Re-ran the corrected main table** (ours = auto, full-N) + verified RAG reproduces. [main table](#)
- **Compared vs *all* baselines** (not just RAG): honest — KV-eviction wins literary-MC, full wins dense-reasoning; we own retrieval/extractive/needle/ultra-long *and* linear. [best-of-all](#)
- **Code review** (methods/benches/eval) + **Occam v3.0** (IMP-lite: one mode, one signal, one knob). [review](#) · [v3.0](#)

This week (2/3) — Paper C: linear-attention line opened

- **Research line + references** for linear attention & variants (Qwen3.5/3.6 GDN family only; no Qwen3 control). Landscape: delta-rule = “memory as online least-squares”; frontier = **how to erase vs write** the compressed state (GDN-2’s channel-wise erase gate carries most of its gain). [references](#) · [ablation design](#)
- **GDN anti-forgetting modules (training-free, real benches): input-side compression rescues GDN on ultra-long** (∞ Bench full 55 $\rightarrow$ imp-lite 76); recency-reorder modules are marginal/task-dependent. [module results](#)
- **Honesty:** the synthetic single-needle probe was inconclusive (task too easy / a probe bug we caught + reverted); trustworthy evidence is the real-bench grid. 128k probe OOMs (GDN kernel).

This week (3/3) — reporting deck + housekeeping

- **Research deck** ([main-research](#)): A/B/C overview, indexed, all artifacts GitHub-hyperlinked; total→detail structure with the mainline logic up front.
- **Scale/family directive (Paper C)**: headline at **10–30B**, GDN family (Qwen3.5/3.6) only; secure the in-band **Qwen3.6-27B** checkpoint.
- **Cluster hygiene**: reclaimed a GPU held for 5 days by a stale demo server (mis-attributed as external); 6 clean GPUs now available.

Cross-cutting insights

Long-context = keep-the-gold-span (gold-recall predicts accuracy); no universal compressor; retrieval can *denoise above full*; the diagnosis is model-invariant; KV-eviction owns literary-MC but **can't run on linear** — that gap is ours.

- Paper B** run the IMP-lite Occam ablation; finalize v3.0; settle the “>RAG or RAG-configured” question.
- Paper C** secure Qwen3.6-27B + RULER (HF) access; diagnose GDN forgetting on real long-context; test input-side vs. erase-gate-retrofit fixes.

Plan 08 / Long-Context Compression — July Week 3

Mon 2026-07-20 to Sun 2026-07-26

Status through Thu 2026-07-23 10:59 PT.

Current priority: finish **Paper A** with complete long-context, generality, budget, ablation, cost, and official-baseline evidence.

Paper A — Introduction logic (six paragraphs)

❶ **Why long context matters.**

It is the working state of document, tool, conversation, and agent systems.

❷ **The problem is not only capacity.**

Long inputs cost more, but even a context that fits may be used unreliably.

❸ **Compression is not automatically robust.**

A shorter state may remove the fact or relation required by the next decision.

❹ **The overlooked question is failure detection.**

Average compressed accuracy does not say whether memory is sufficient for this input.

❺ **Agent design should preserve two paths.**

Use compact memory when trusted; let a gate return to raw evidence otherwise.

❻ **GCM implements the interface.**

One frozen backbone builds recurrent memory, reads it, estimates confidence, and provides raw fallback.

Paper focus

Robust long-context compression: map where soft memory works, detect where it does not, and measure how much a shared-backbone fallback recovers.

Paper A — story and completed core evidence

Story

Gated Context Memory (GCM) treats soft memory as a **routed interface**: use a recurrent short state when it is sufficient, and preserve a bounded raw path when the needed evidence does not survive compression.

base	method	QuALITY	BFCL	HotpotQA
Qwen3-8B	Raw ref.	−†	92.4	53.7
	LLMLingua-2	−†	70.3	22.1
	Compressor (w/o gate)	−†	72.3	28.9
	Compressor (w/ gate)	−†	88.5	50.9
Qwen3.5-9B	Raw ref.	−†	84.5	53.9
	LLMLingua-2	−†	60.8	28.9
	Compressor (w/o gate)	−†	72.0	30.5
	Compressor (w/ gate)	−†	80.5	51.7

- †QuALITY is quarantined after a zero-based label-index bug; BFCL/Hotpot remain valid.
- Formal fixed-family test certifies no nonzero coverage; routing remains held-out empirical analysis.

Paper A — real long-context result

base	method	LongBench-v2	∞ Bench-choice	BABILong
Qwen3-8B	Raw ref.	31.2	52.8	18.1
	Compressor (w/o gate)	— [†]	— [†]	— [‡]
	Compressor (w/ gate)	— [†]	— [†]	18.1
Qwen3.5-9B	Raw ref.	33.0	52.4	10.5
	Compressor (w/o gate)	— [†]	— [†]	— [‡]
	Compressor (w/ gate)	— [†]	— [†]	11.9

Finding

On multi-hop transfer, the compressor without the gate can beat truncated raw (Qwen3 HotpotQA 19.1 $\rightarrow$ 30.5; Qwen3.5 3.9 $\rightarrow$ 26.6). On single-document / extractive targets, the gated variant falls back. Across the harvested two-base targets, **Compressor (w/ gate) $\geq$ Raw in every cell.**

[†]QuALITY-source invalid; [‡]Hotpot $\rightarrow$ BABILong collapse. [all-benchmark tables](#)

Paper A — full status and execution order

stage	done	run	fail	pending	Live
core main	60 valid	0	28 invalid	0	0
replicate + SFT audit	0 valid	0	9 invalid	0	0
transfer adapters	27 valid	0	16 invalid	0	0
real long-context	81 valid	0	25 affected	0	0
7-model generality	13 valid	2 invalid	8	0	0
budget / RULER	0 valid	1 window	2 fail	18	1
ablation	0	0	0	36	2
official baselines	0	0	0	52	3

Next

- 1 stop stale/OOM workers;
- 2 sync corrected loader and code;
- 3 validate one clean QuALITY cell;
- 4 rerun quarantined cells;
- 5 repair RULER, then ablation/baselines.

benchmark + baseline inventory experiment receipt